



# Northern University of Business & Technology Khulna

Department of Computer Science and Engineering

Course Code: CSE 2110

Course Title: Object Oriented Programming Lab

## PROJECT PROPOSAL

**Project Name: Smart Student Routine & Expense Manager**

| Submitted To                                                                                                                                            | Submitted By                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
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# 1. Project Overview

## Background

University life presents a dual challenge for students: managing complex academic schedules and maintaining a tight personal budget. Without a centralized system, students often miss deadlines or overspend.

## Problem Statement

Existing solutions for routine management and expense tracking are often separate, requiring users to switch between multiple platforms, which leads to fragmented data and inefficiency.

## Existing Problems

- Manual tracking of expenses is prone to errors.
- Paper-based schedules are easily lost or not updated in real-time.
- Lack of integrated insights linking time management with financial habits.

## Proposed Solution

The "Smart Student Routine & Expense Manager" is a unified desktop application built on Object-Oriented Programming (OOP) principles. It offers a seamless interface to manage daily academic tasks alongside a comprehensive financial ledger, featuring secure authentication using password hashing with PBKDF2WithHmacSHA256 and robust error handling for invalid input, database errors, empty fields, and duplicate data.

## Scope

The project focuses on creating a local desktop application that handles routine scheduling, automated reminders, expense logging, and basic financial reporting for students.

## Significance

By integrating these two critical aspects of student life, the application fosters better discipline and financial literacy, ultimately reducing academic and personal stress.

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## 2. Objectives

### Primary Objective

- To develop a robust, user friendly application that integrates schedule management and expense tracking into a single interface using Java and OOP, ensuring secure authentication via PBKDF2WithHmacSHA256 and comprehensive error handling.

### Secondary Objectives

- Implement secure user authentication.
- Enable automated notifications for upcoming classes and exams.
- Provide visual summaries of monthly spending vs. budget.

## 3. Literature Review

The following table compares the proposed system with existing market solutions:

| Feature       | Physical Diary | Generic Expense Apps | Proposed Smart Manager                        |
|---------------|----------------|----------------------|-----------------------------------------------|
| Integration   | None           | Partial ▾            | High (Routine + Finance)                      |
| Reminders     | Manual         | Automated ▾          | Automated (Context-aware)                     |
| Data Security | Low            | High ▾               | Medium, local database with password hashing. |
| Portability   | Medium         | High ▾               | High (Desktop/Portable)                       |

## 4. Functional Requirements

- **User Authentication:** Secure login and registration.
  - **Routine Management:** Create, edit, and delete class/exam schedules. Add: Reminder, ReminderService, NotificationManager. The application will include in-app notifications (e.g., 'Class starts in 30 minutes').
  - **Expense Tracker:** Log daily transactions with category tags (food, transport, etc.).
  - **Budgeting:** Set monthly limits and receive warnings.
  - **Reporting:** Generate visual reports of spending (Monthly expense summary, Category breakdown, Budget comparison).
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## 5. Non-Functional Requirements

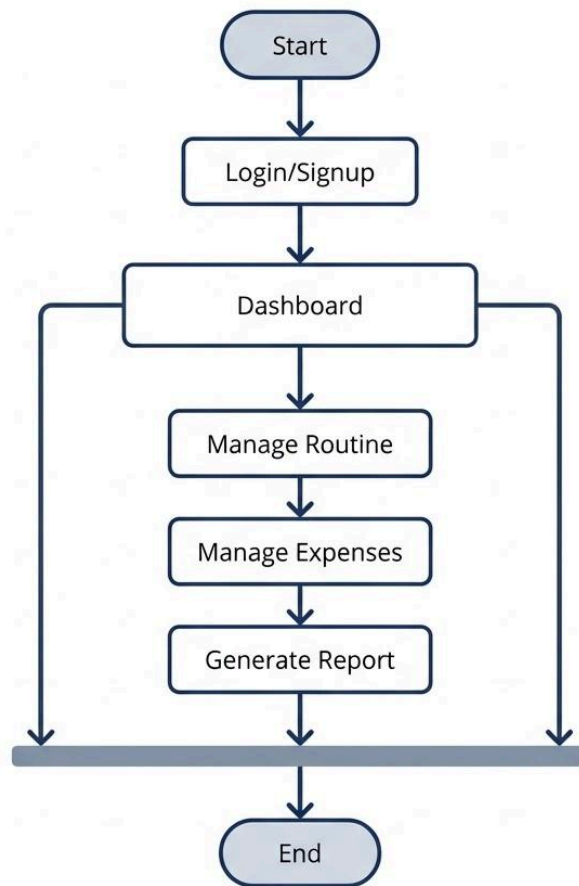
- **Usability:** Simple UI designed for quick data entry.
  - **Performance:** Fast response time for searching routines.
  - **Reliability:** Data persistence to ensure logs are never lost.
  - **Maintainability:** Modular code structure for future updates.
  - The project will demonstrate the following OOP concepts: Encapsulation, Inheritance, Polymorphism, Abstraction, Association, Composition, Method Overloading, Method Overriding, Interfaces, and Exception Handling.
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## 6. Methodology

The project follows the **Software Development Life Cycle (SDLC)** Waterfall model to ensure structured progress:

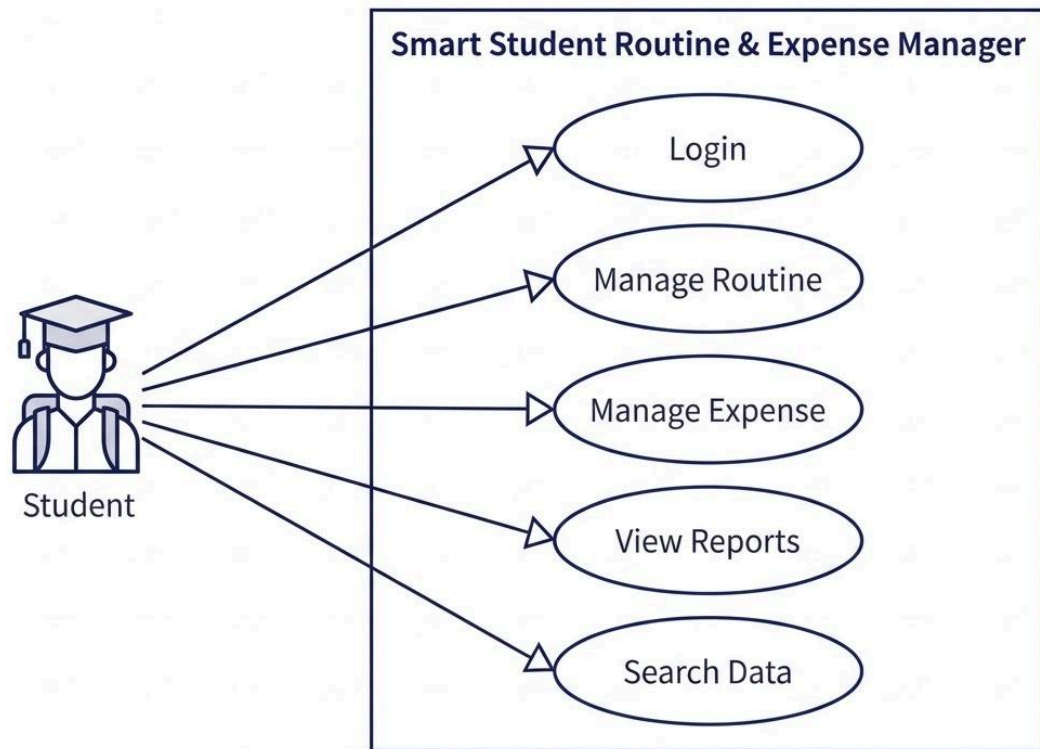
1. **Requirement Analysis:** Gathering student needs.
  2. **System Design:** Creating UML diagrams and database schema.
  3. **Implementation:** Coding using Java (OOP).
  4. **Testing:** Unit testing and integration testing.
  5. **Deployment:** Final packaging of the application.
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## 7. Professional Flowchart

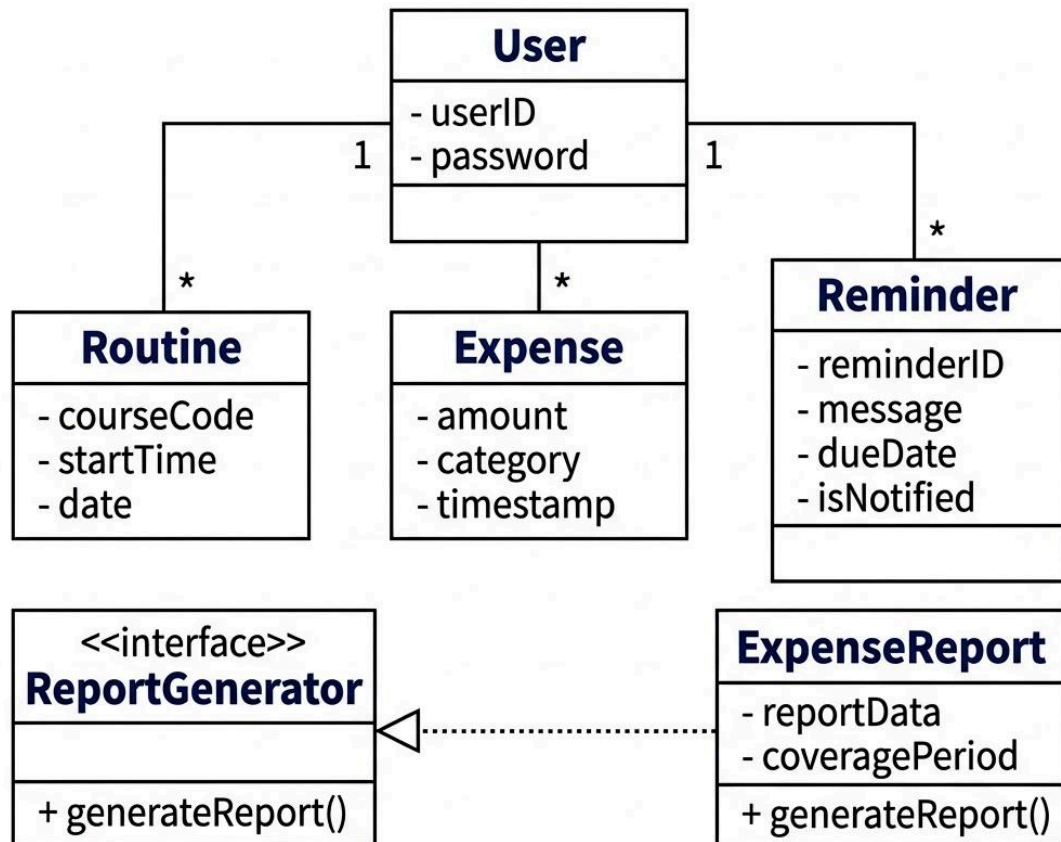




## 8. UML Use Case Diagram

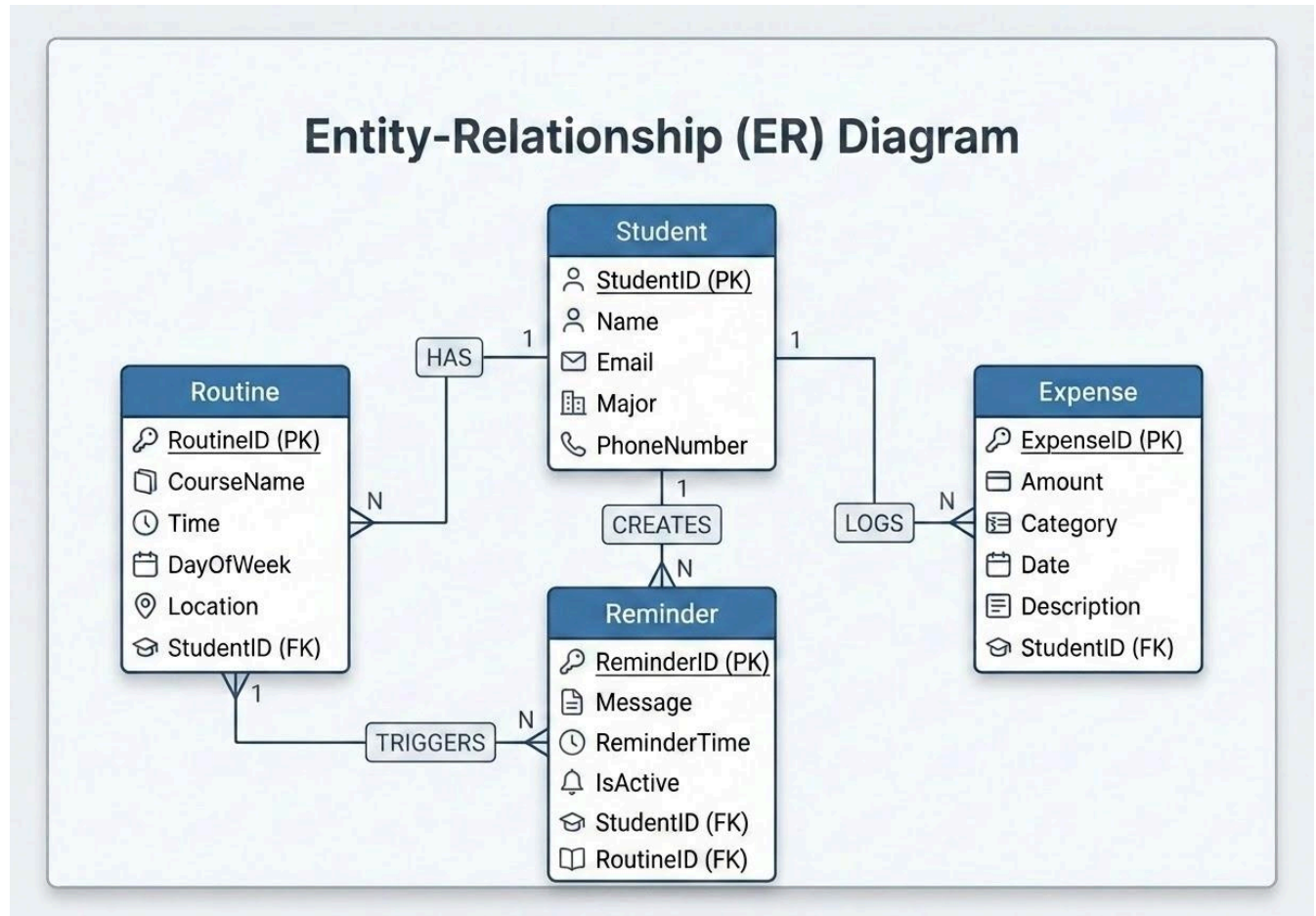


## 9. Class Diagram

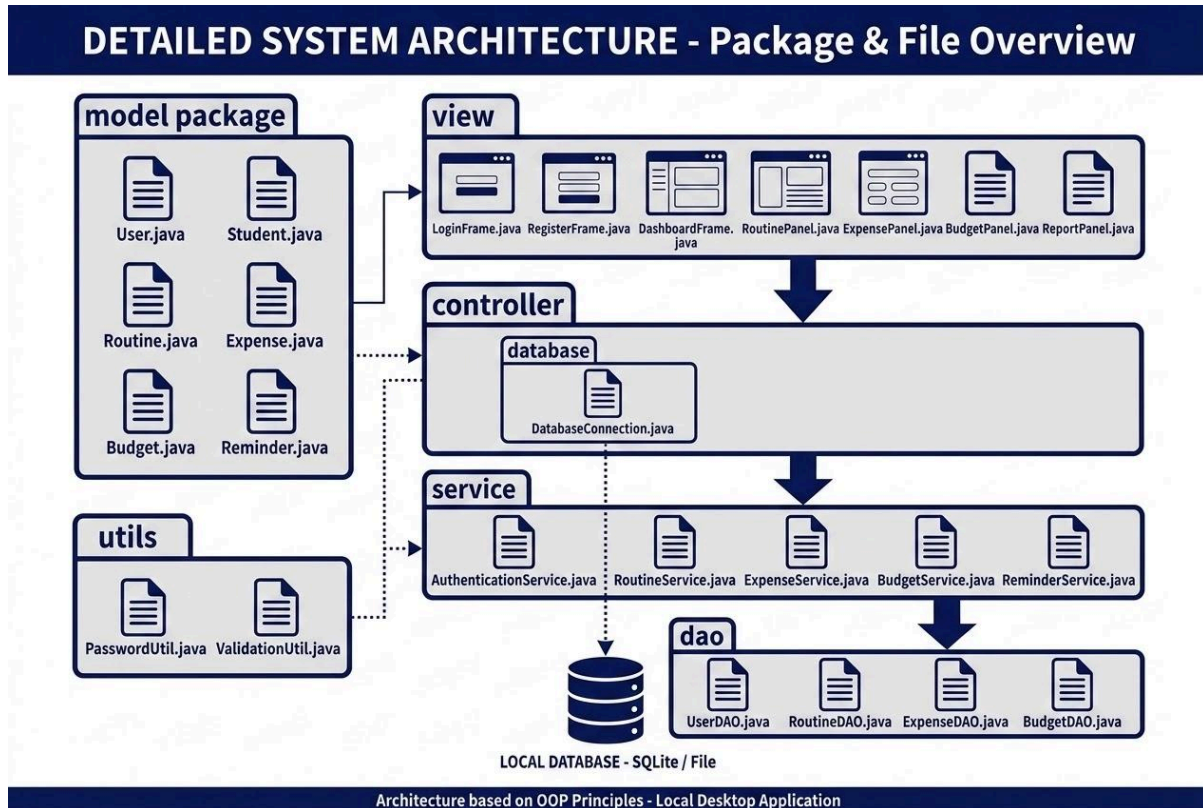




## 10. ER Diagram



# 11. System Architecture



# 12. Gantt Chart

| Phase                | D1 | D2 | D3 | D4 | D5 | D6 | D7 | D8 | D9 | D10 |
|----------------------|----|----|----|----|----|----|----|----|----|-----|
| Requirement Analysis | ✓  | ✓  |    |    |    |    |    |    |    |     |
| System Design        |    | ✓  | ✓  |    |    |    |    |    |    |     |

| Phase                 | D1 | D2 | D3 | D4 | D5 | D6 | D7 | D8 | D9 | D10 |
|-----------------------|----|----|----|----|----|----|----|----|----|-----|
| Database Design       |    |    | ✓  | ✓  |    |    |    |    |    |     |
| OOP Model Development |    |    |    |    | ✓  | ✓  |    |    |    |     |
| Authentication        |    |    |    |    |    | ✓  |    |    |    |     |
| Routine Module        |    |    |    |    |    |    | ✓  | ✓  |    |     |
| Expense Module        |    |    |    |    |    |    |    | ✓  | ✓  |     |
| Budget Module         |    |    |    |    |    |    |    |    | ✓  | ✓   |
| Reports               |    |    |    |    |    |    |    |    |    | ✓   |
| Testing               |    |    |    |    |    |    |    |    | ✓  | ✓   |
| Documentation         |    |    |    |    |    |    |    |    | ✓  | ✓   |

## 13. Expected Outcomes

- A fully functional Java application.
- A cleaner organization of student academic schedules.
- Improved financial awareness through detailed expense logs.

## 14. Resources Required

### Software

- **Language:** Java (JDK 17+)
- **IDE:** IntelliJ IDEA or Eclipse
- **Database:** SQLite
- **Tools:** Draw.io for diagrams

## Hardware

- Minimum 8GB RAM Laptop
- Core i3 Processor or higher

## 15. Future Scope

- Integration of AI to suggest budget optimizations.
- Syncing capabilities with Google Calendar.
- Development of a companion mobile application.

## 16. Conclusion

The Smart Student Routine & Expense Manager aims to address the common struggles of university life. By leveraging OOP principles, the system will provide a structured, efficient, and reliable way for students to regain control over their time and finances.